ARCHITECTURE & UX REVIEW • READ ONLY

ProLingo v6 Beta — Structured Text UX Audit

Comprehensive Evaluation of Player Settings, Audio Sidebar, Paragraph Card Expansion & Mobile Ergonomics

REPOSITORY	BRANCH / BASELINE	DATE & TIME	AUDIT STATUS
nazurinian/prolingo-app	v6-beta / v6.0.3-beta.8-p7-hf3	2026-11-10 12:00 WIB	COMPLETE • 100% Read-Only

1 Executive Summary & Audit Baseline

ProLingo v6 Beta has established an advanced audio playback architecture featuring Split/Full dual-representation modes, dual-channel audio sequencing (English Text and Indonesian Meaning), IndexedDB binary staging, Portable ZIP archives, and hierarchical speaker voice overrides. All existing P9-A automated runtime suites (Split/Full modes, Pause/Resume/Stop, Local First with TTS fallback, and localStorage persistence) pass successfully.

The goal of this independent audit is to identify usability bottlenecks, confusing terminology, nested dropdowns, and card expansion defects, providing an actionable redesign that makes the existing controls intuitive and ergonomic on both desktop and mobile—without modifying or replacing the audio engine.

📊 DIAGRAM 1: CURRENT UX BOTTLENECKS VS PROPOSED TARGET ARCHITECTURE

⚠ CURRENT BOTTLENECKS		✨ PROPOSED ARCHITECTURE
5-Box Stacked Modal: Excessive vertical height & micro-typography (text-[7px])	→	**Tabbed / Segmented Settings:** Fast 1-tap pills, no dropdowns for <= 3 choices
Card Collapse on Stop: Expanded reading state abruptly snaps shut on pause/stop	→	**3 Card Expansion Modes:** Follow Playback, Expand All, Custom + The Stop Rule
Full Mode Active Blindness: playingIndex=null leaves active card unhighlighted	→	**Active Card in Full Mode:** Track activeBlockId so cards highlight & follow
Terminology Fragmentation: Local Allowed vs Local First vs Auto vs Custom	→	**Clean 3-Layer Audio IA:** Strict separation of Heard, Generated, and Stored
Hidden Headings Quick-Play: No single-tap EN-only or ID-only play on card text	→	**Text & Meaning Headings:** [▶ Play EN] & [▶ Play ID] single-tap buttons

Key Audit Invariant: All recommendations strictly respect the HF2 architectural invariants: single global TTS Only source of truth, global local voice fallback priority, independent manual playback controls, separate Edge generation/download defaults, and strict backward compatibility with existing IndexedDB and Text DB v1 data contracts.

2 Settings Inventory & True Engine Function Analysis

A rigorous audit of the source files (`TextStructuredPlaybackControls.jsx`, `TextStructuredPlayer.jsx`, `TextStructuredAudioControls.jsx`, and `textStructuredPlaybackPreferenceDomain.js`) revealed that several settings

perform functions quite different from what their visual UI labels imply:

SETTING IDENTIFIER	VISIBLE LABEL & LOCATION	TRUE ENGINE FUNCTION	OBSERVED UX DISCREPANCY
`playbackRepresentationMode`	Segmented button: `SPLIT` / `FULL` (Player Settings & Bottom Bar)	Controls audio chunking & reading granularity. Split uses sentence-level segments and visual tracking. Full plays uninterrupted continuous card audio.	Located inside a container titled "Playback Audio", misleading users into thinking it configures audio output devices rather than content representation.
`audioSourceMode` / `ttsOnly`	Button: `TTS ONLY • ON/OFF` (Settings) `LOCAL FIRST / TTS ONLY` (Bottom Bar)	Global bypass: forces the browser Web Speech API for all documents, completely bypassing IndexedDB Staging and Portable ZIPs without deleting files.	Presents 3 conflicting labels: "LOCAL ALLOWED" in Audio Sidebar, "LOCAL FIRST" in Bottom Player, and "TTS ONLY • OFF — local audio allowed" in Player Settings.
`globalAudioPlaybackOrderV1`	Reorder list + Add button (Player Settings)	Defines multi-voice fallback sequence (e.g. `Sonia > Libby`). The first voice with a `READY` local binary in Staging is played.	Placed inside Player Settings rather than Audio Sidebar, disconnecting voice priority from voice generation and inventory inspection.
`displayMode`	4 pills: `EN only`, `EN + active ID`, `EN + ID`, `ID only`	Visual card filtering: conditionally hides or displays the English text block and Indonesian translation block.	Clear function, but placed directly next to "Global EN / ID order", causing users to confuse visual text display with audio playback channels.
`playbackChannelMode`	4 pills: `EN`, `EN → ID`, `ID → EN`, `ID`	Audio sequence per segment during Document Play: plays EN only, ID only, or alternating bilingual channels.	Separated from other Global playback behaviour (order, repeat, delays) by the "Manual Play" block.
`manualSegmentPlaybackChannelMode` & `manualSegmentRepeatMode`	2 Dropdowns: "Channels" & "Repeat" (Player Settings)	Audio channels and repetition used when clicking "Only" on a segment row or tapping a sentence in Split reading mode.	No "(Follow Global)" option. Uses native dropdowns for only 4 choices, forcing 2-3 clicks to change.
`manualCardPlaybackChannelMode` & `manualCardRepeatMode`	2 Dropdowns: "Channels" & "Repeat" (Player Settings)	Audio channels and repetition used when clicking "Card" on a card header.	Duplicates the Segment Only controls identically, creating a dense wall of 4 native select boxes.
`playbackOrderMode`	Dropdown: `Sequential` / `Shuffle`	Segment traversal sequence during Document Playback.	Uses a native dropdown for an exclusively binary (2-state) choice.
`repeatMode`	Dropdown: `Once` / `2×` / `Loop`	Document loop iteration count.	Uses a native dropdown for only 3 static options.
`channelDelayMs` & `segmentDelayMs`	Number inputs: `EN/ID gap` & `Segment gap`	Pause duration (ms) inserted between channels (EN/ID) and between consecutive segments.	Raw millisecond numeric input with tiny stepper arrows; awkward on mobile without quick presets (0.5s, 1s, 2s).

SETTING IDENTIFIER	VISIBLE LABEL & LOCATION	TRUE ENGINE FUNCTION	OBSERVED UX DISCREPANCY
`resumeMode`	Dropdown: `Continue` / `Restart`	Determines playback index after Stop (resume from last segment vs restart from Card 1).	Uses a native dropdown for a binary choice.

3 Verified UX Problems & Cognitive Friction Points

3.1. Sudden Card Collapse on Stop (Critical Defect)

> **Source-Code Verified Defect:** In `TextStructuredPlayer.jsx` (line 159):
> `const expanded = manualExpanded || isActiveCard || isFocusCard;`
> When playback is active, `isActiveCard` is true, auto-expanding the card. When the user hits **Stop** (or when playback finishes the last segment and stops), `isActiveCard` immediately switches to `false`. If the user did not explicitly toggle the card's chevron (meaning `manualExpanded === false`), the card **snaps shut instantly**.

Impact: The learner is abruptly ejected from their visual reading context the moment audio stops, losing their place in the paragraph.

3.2. Active Card Indicator Blindness in FULL Representation Mode

> **Source-Code Verified Defect:** In `textStructuredPlaybackSessionService.js` (lines 264–266):
> `const suppressHighlight = normalizedRepresentation === TEXT_STRUCTURED_PLAYBACK_REPRESENTATION_MODES.FULL;`
> `setPlayingIndex(suppressHighlight ? null : item.id);`
> `setCurrentIndex(item.id);`
> In Full mode, `playingIndex` is forced to `null` to suppress sentence-level highlighting. However, `StructuredPlayerCard` determines its active status via `segments.find(s => s.id === playingIndex)`. Because `playingIndex` is `null`, `isActiveCard` evaluates to `false` for every single card in the document!

Impact: During Full playback, the currently playing card receives no active border, displays no `ACTIVE` badge, never auto-expands, and fails to auto-scroll into view.

3.3. Disconnected Heading Quick-Play Controls

Inside an expanded Paragraph card (`TextParagraphUnifiedReading.jsx`), the Text (English) and Meaning (Indonesian) blocks have plain uppercase text headings: `<p>Text</p>` and `<p>Meaning</p>` without any action buttons. To listen to only the English paragraph, or only Indonesian, a user must currently change their global or manual preferences in Player Settings, or switch on Split mode and click sentences individually.

3.4. Overuse of Native Dropdowns & Micro-Typography

`TextStructuredPlaybackControls.jsx` uses heavy micro-typography (`text-[7px]`, `text-[8px]`, `text-[9px]`). Options with only 2 to 4 choices (Sequential/Shuffle, Continue/Restart, Repeat, Channels) are implemented as native `<select>` dropdowns, creating high tap friction and visual clutter.

4 Duplicated, Redundant & Confusing UI Surfaces

CONTROL / CONCEPT	SURFACES WHERE IT CURRENTLY APPEARS	UX RECOMMENDATION
Split vs Full Mode	1. `BottomPlayerBar` (Toggle button) 2. `TextStructuredPlaybackControls` (Button grid)	Keep on Bottom Bar for quick toggle; in Player Settings, group under "Reading Mode", not "Playback Audio".
TTS Only Global Bypass	1. `BottomPlayerBar` (LOCAL FIRST / TTS ONLY) 2. `TextStructuredPlaybackControls` (ON/OFF btn) 3. `TextStructuredAudioControls` (Status banner)	Harmonize labeling to **LOCAL FIRST** / **TTS ONLY** across all surfaces. Preserve single source of truth.
Local Voice Priority Order	1. `BottomPlayerBar` (Summary string) 2. `TextStructuredPlaybackControls` (Reorder list) 3. `TextStructuredCardAudioPanel` (Card override)	Move or co-locate priority ordering under Audio Sidebar "Playback" tab with a shortcut in Player Settings.
Audio Channel Selectors	1. Player Settings: Global EN/ID order 2. Player Settings: Manual Segment Channels 3. Player Settings: Manual Card Channels	Add explicit "(Follow Global)" default option to Manual Card and Segment dropdowns to show inheritance.
Readiness Metrics	1. Workspace Header (Compact pills) 2. Card Header (Compact pills) 3. Card Audio Panel (Coverage grid) 4. Audio Sidebar (TextAudioDataPanel tiles)	Maintain compact counters on Workspace/Card headers; clarify "History*" definition via an inline tooltip.

5 Recommended Information Architecture

To resolve confusion between what is heard, what is synthesized, and what is stored, we define a clear **3-Pillar Information Architecture**:

🏛️ **DIAGRAM 2: THE THREE PILLARS OF PROLINGO AUDIO ARCHITECTURE**

1. WHAT IS PLAYED

Playback Engine (Live Output):

- **Representation:** Split vs Full
- **Channels:** EN, EN→ID, ID→EN, ID
- **Routing:** Local First vs TTS Only
- **Local Priority:** Sonia > Ryan > Libby
- **Live Browser TTS:** Female UK default, independent rates
- **Feel:** Delays, Repeat, Order, Resume

2. WHAT IS GENERATED

Edge Synthesis (Background):

- **Service Health:** Edge WebSocket check
- **Global Defaults:** EN (Sonia), ID (Tuti)
- **Synthesis Tuning:** Rate (±50%), Pitch (±20Hz)
- **Presets:** Default / Follow Player / Custom
- **Actions:** Generate Missing, Regen Card
- *Never changes live playback voice*

3. WHAT IS STORED

Storage & Assets (IndexedDB):

- **Staging (IndexedDB):** Binary MP3/WAV blobs
- **Portable ZIP:** Package import & validation
- **Readiness Status:** Ready, History*, Stale, Missing
- **Storage Actions:** Clear Staging, Clear Cache
- **Folder:** Deprecated & locked in beta.8

Edge Synthesis ➔ Generates MP3s into ➔ IndexedDB Staging | **Portable ZIP ➔ Merges Blobs into ➔ IndexedDB Staging** | **IndexedDB Staging ➔ Feeds Ready Audio to ➔ Playback Engine**

6 Paragraph Card Expansion & Active-Card Architecture

6.1. Three First-Class Expansion Modes

We introduce an explicit `cardExpansionMode` preference with three clear operating modes:

🔄 **DIAGRAM 3: CARD EXPANSION STATE TRANSITION & PERSISTENCE RULES**

DEFAULT

Follow Playback

Active card expands automatically as audio plays

READING

Expand All

All cards opened for seamless continuous reading

MANUAL

Custom

Expansion governed exclusively by user clicks

The Stop Rule: When playback stops (via Stop button, Pause, or end of list), the currently active card **MUST REMAIN EXPANDED**. It never snaps shut unexpectedly.

The Manual Transition Rule: If the user manually clicks any card's chevron while in `follow-playback` mode, the system automatically transitions the document to `custom` mode so the user's manual intent is preserved.

6.2. Active Card Resolution in FULL Mode

To fix the bug where Full mode renders cards "blind", `TextStructuredPlayer.jsx` will track `activeBlockId` at the player level, decoupling card active detection from the sentence-level `playingIndex`:

```
const isActiveCard = playingContext === TEXT_STRUCTURED_PLAYBACK_CONTEXT
  && (isPlaying || isPaused)
  && (block.segments?.some(s => s.id === playingIndex) || block.id === activeBlockId);
```

Sentence-level background highlighting remains suppressed in Full mode as required by P9-A, while the **card** properly receives its indigo border, `ACTIVE` badge, and auto-scroll behavior.

6.3. Quick-Play Controls on TEXT & MEANING Headings

In `TextParagraphUnifiedReading.jsx`, the TEXT and MEANING block headers are upgraded with single-tap quick-play buttons:

```
┌──────────────────────────────────────────────────────────────────┐
│ TEXT [▶ Play EN] [Copy] │
│ The mysterious traveler arrived at the ancient village as dusk fell... │
├──────────────────────────────────────────────────────────────────┤
│ MEANING [▶ Play ID] [Copy] │
│ Pengelana misterius itu tiba di desa kuno saat senja mulai turun... │
└──────────────────────────────────────────────────────────────────┘
```

7 Global, Manual, and Voice Inheritance Model

DIAGRAM 4: AUDIO CHANNEL & VOICE OVERRIDE CASCADE

CHANNEL & REPEAT INHERITANCE:

1. **Global Playback:** Sets document sequence (EN/ID order) & repeat.
2. **Card Play ('Card' button):** Defaults to "(Follow Global)" or uses explicit card channel mode. Always sequential within the card.
3. **Segment Only ('Only' button):** Defaults to "(Follow Global)" or uses explicit segment channel mode. Always Split representation.
4. **Heading Quick-Play:** Overrides channel to `TEXT_ONLY` or `MEANING_ONLY` for a single run.

VOICE RESOLUTION CASCADE:

1. **Segment Override** (if explicit voice assigned)
2. **Speaker Assignment** (in conversation cards)
3. **Card Override** (card-specific voice assignment)
4. **Document Override** (workspace voice assignment)
5. **Global Browser TTS Default** (Web Speech API)

8 Desktop and Mobile Wireframes & Control Recommendations

8.1. Streamlined Player Settings (Tabbed Modal Dialog)

Consolidates 5 stacked boxes into 3 logical tabs, replacing native dropdowns with instant segmented buttons:

```
┌──────────────────────────────────────────────────────────────────────┐
│ ⚙ Text Player Settings [X] │
├──────────────────────────────────────────────────────────────────────┤
│ [ General ] [ Manual Play ] [ Voice Priority ] │
├──────────────────────────────────────────────────────────────────────┤
│ │
│ DISPLAY ON CARDS │
│ [ EN only ] [ EN + active ID ] [ (●) EN + ID ] [ ID only ] │
│ │
│ GLOBAL PLAYBACK CHANNELS │
│ [ (●) EN ] [ EN → ID ] [ ID → EN ] [ ID ] │
│ │
│ PLAYBACK BEHAVIOUR │
│ Order: [ (●) Sequential ] [ Shuffle ] │
│ Repeat: [ (●) Once ] [ 2× ] [ Loop ] │
│ On Stop: [ (●) Continue from last ] [ Restart from beginning ] │
│ │
│ DELAYS & PAUSES │
│ EN / ID Gap: [ None (0s) ] [ (●) 0.5s ] [ 1.0s ] [ 1.5s ] [Custom]│
│ Sentence Gap: [ None (0s) ] [ 0.5s ] [ (●) 1.0s ] [ 2.0s ] [Custom]│
│ │
└──────────────────────────────────────────────────────────────────────┘
```

8.2. Mobile Bottom Player Bar (Clean Ergonomic Touch Layout)

Replaces the squeezed 1-row mobile layout with a stacked 2-row design providing minimum 44px touch targets:

```
┌──────────────────────────────────────────────────────────────────────┐
│ NOW PLAYING │
│ 1. Chapter 1: The Departure • EN • "The morning sun..." (1/12) │
├──────────────────────────────────────────────────────────────────────┤
│ [⏮ Prev] [ ▶ PLAY ] [⏭ Next] [■ Stop] │
├──────────────────────────────────────────────────────────────────────┤
│ [ Split / Full ] [ Local / TTS ] [ ⚙ Settings ] │
└──────────────────────────────────────────────────────────────────────┘
```

9 Prioritized Recommendations Matrix

ID	RECOMMENDATION DESCRIPTION	CLASSIFICATION	PRIORITY	P9-A SAFETY FLAG
REC-01	**Fix Card Collapse on Stop:** Keep active card expanded when playback stops; never snap shut.	State / UI	**P0 (CRITICAL)**	🟢 **SAFE** Purely view state; audio engine untouched.
REC-02	**Restore Active Card in Full Mode:** Highlight and follow active card during Full playback.	State / UI	**P0 (CRITICAL)**	🟡 **WATCH** Keep `setPlayingIndex(null)` sentence suppression intact.
REC-03	**Add 3 Card Expansion Modes:** Expose Follow Playback, Expand All, and Custom in toolbar.	State / UI	**P1 (HIGH)**	🟢 **SAFE** Backward compatible preference addition.
REC-04	**Heading Quick-Play Buttons:** Add [▶ Play EN] and [▶ Play ID] on paragraph block headings.	UI Only	**P1 (HIGH)**	🟢 **SAFE** Uses existing card playback dispatchers.
REC-05	**Segmented Buttons for Dropdowns:** Replace 2-3 option dropdowns with 1-tap segmented pills.	UI Only	**P1 (HIGH)**	🟢 **SAFE** Normalizes to identical preference values.
REC-06	**"Follow Global" Manual Option:** Explicitly indicate inheritance on Manual Channel selectors.	UI / Copy	**P2 (MED)**	🟢 **SAFE** Aligns UI with existing domain fallback.
REC-07	**Standardize Delay Presets:** Add 0s, 0.5s, 1s, 2s preset pills alongside numeric custom input.	UI Only	**P2 (MED)**	🟢 **SAFE** Output values remain millisecond numbers.
REC-08	**Streamline Audio Sidebar:** Partition clearly into Playback, Generation, and Storage.	UI / Copy	**P2 (MED)**	🟢 **SAFE** Preserves existing handlers and DOM IDs.
REC-09	**Mobile Bottom Bar 2-Row Layout:** Stack transport controls above toggles for 44px tap targets.	UI Only	**P2 (MED)**	🟡 **WATCH** Keep `data-text-global-player-controls` test tag.
REC-10	**Clarify History* Status Copy:** Add tooltip explaining binary absence from Staging.	Copy Only	**P3 (LOW)**	🟢 **SAFE** Pure copy clarification.

10 Implementation Plan & Regression Testing

Implementation should proceed in 3 disciplined, incremental phases:

1. **Phase 1 (Card Expansion & Full Mode Fix):** Update `TextStructuredPlayer.jsx` to implement `cardExpansionMode`, maintain card expansion on Stop, and track `activeBlockId` during Full mode. Verify via `audit-v603-beta8-p3-runtime-playback.mjs`.
2. **Phase 2 (Paragraph Heading Quick-Play & Player Settings UI):** Update `TextParagraphUnifiedReading.jsx` to add quick-play buttons, and convert `TextStructuredPlaybackControls.jsx` dropdowns to segmented pills.
3. **Phase 3 (Mobile Player Bar & Audio Sidebar Polish):** Restructure mobile BottomPlayerBar touch targets and clarify Audio Sidebar tab partitioning.

Audit Script Protection: When executing any future UI modifications, the following exact test strings and attributes must be preserved intact to prevent breaking `audit-v603-beta8-p7-hf2` and `audit-v603-beta8-p7-ui-simplification`: `data-text-global-player-controls="true"`, `data-text-global-tts-only="true"`, `data-text-global-playback-mode="true"`, `"Toggle global Split / Full"`, `"Toggle global TTS Only"`, `"LOCAL FIRST"`, `"TTS ONLY"`, and `"first Ready wins"`.

11 Evidence Ledger

- **Source-Code Findings:** Directly verified in `TextStructuredPlayer.jsx` (line 159), `textStructuredPlaybackSessionService.js` (lines 264–266), `TextParagraphUnifiedReading.jsx` (lines 135–175), and `TextStructuredPlaybackControls.jsx` (lines 77–88, 148–153).
- **UX Hypotheses:** Grounded in standard Nielsen-Norman Group usability heuristics (recognition over recall, error prevention, touch target sizing ≥ 44px, immediate affordance for bilingual learning).
- **Automated Test Evidence:** Ran `node scripts/audit-v603-beta8-p7-hf3-mobile-ui.mjs` (6/6 pass), `node scripts/audit-v603-beta8-p7-hf2-global-player-defaults.mjs` (19/19 functional pass), and `node scripts/audit-v603-beta8-p7-ui-simplification.mjs` (16/16 functional pass).